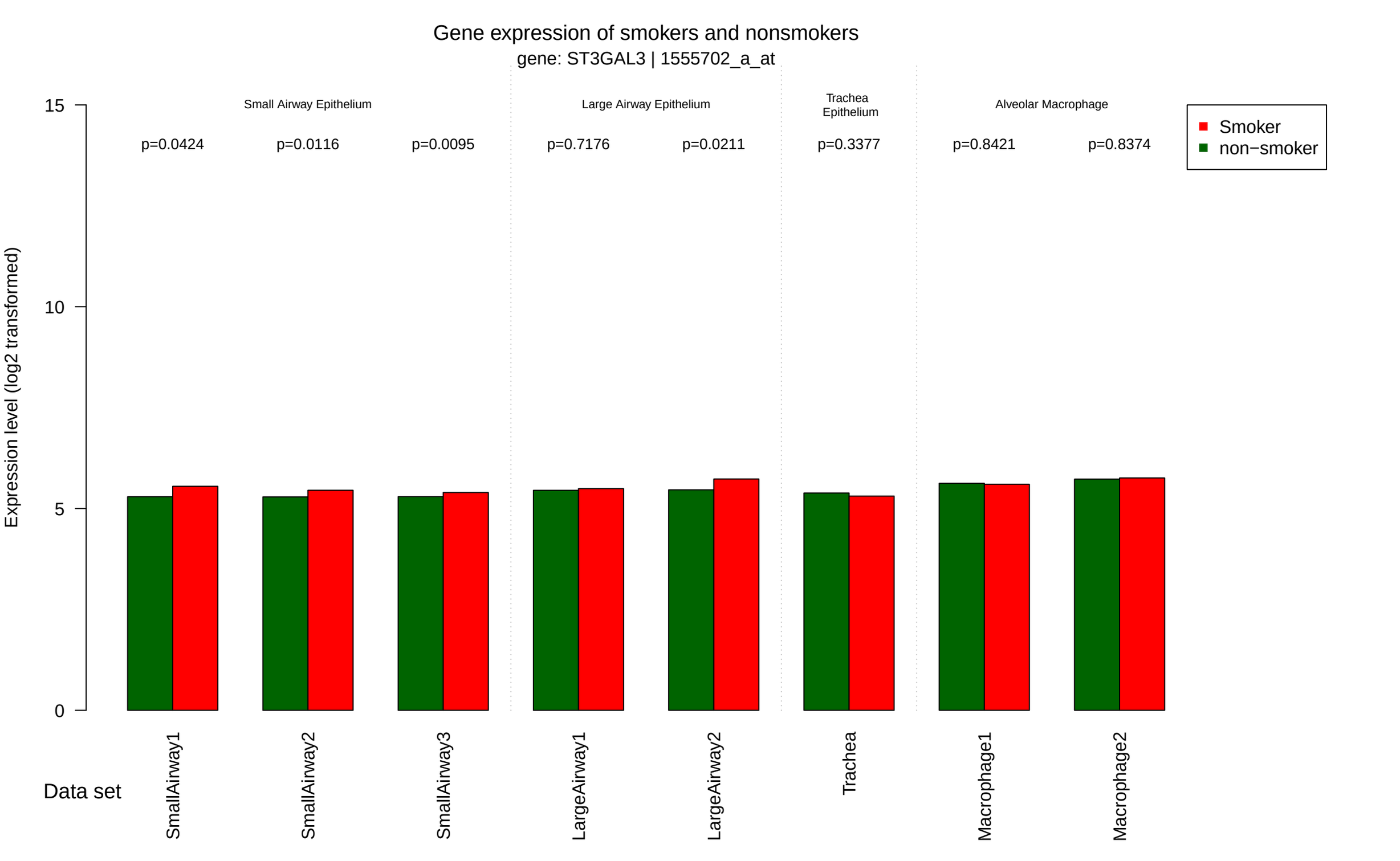
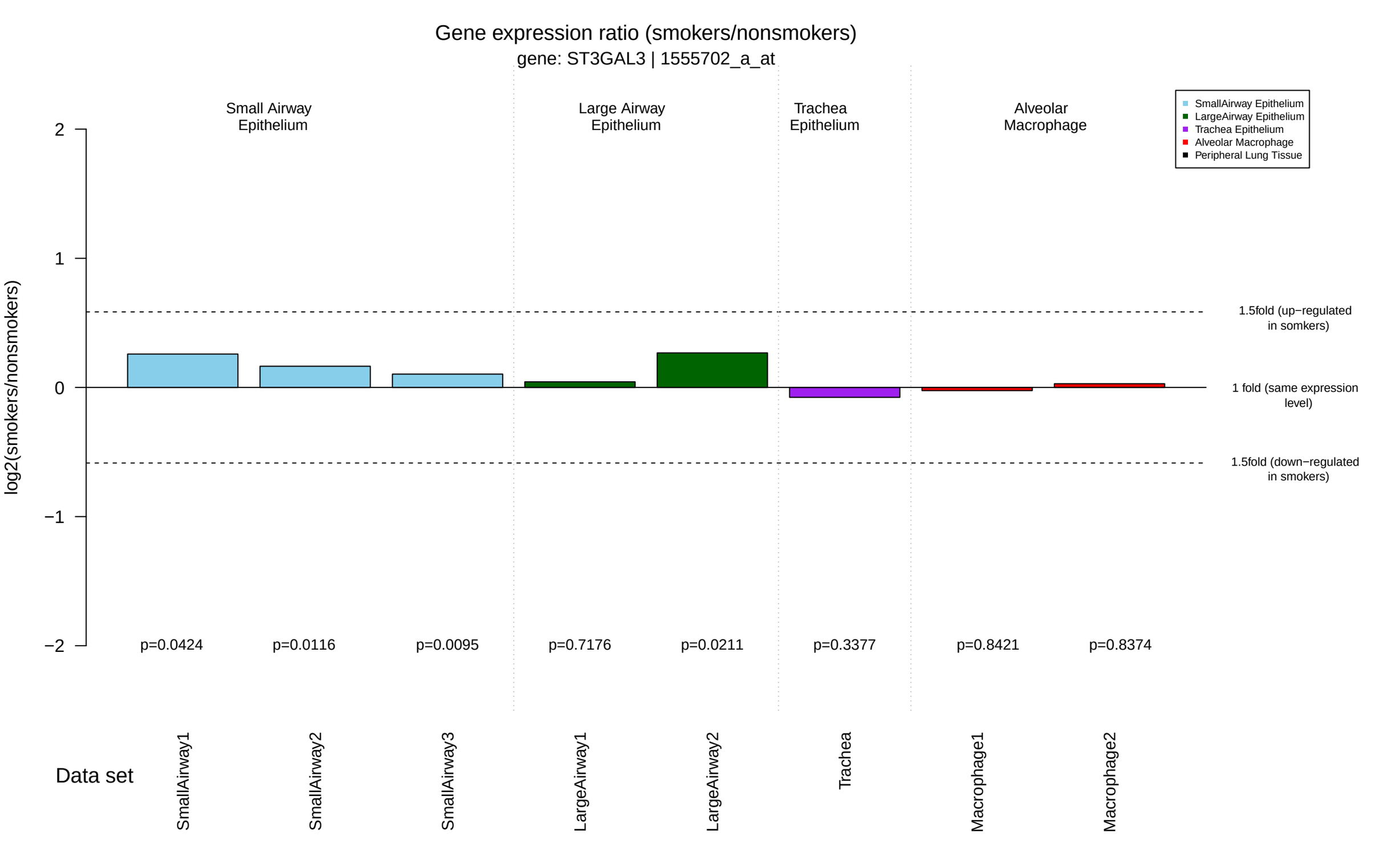
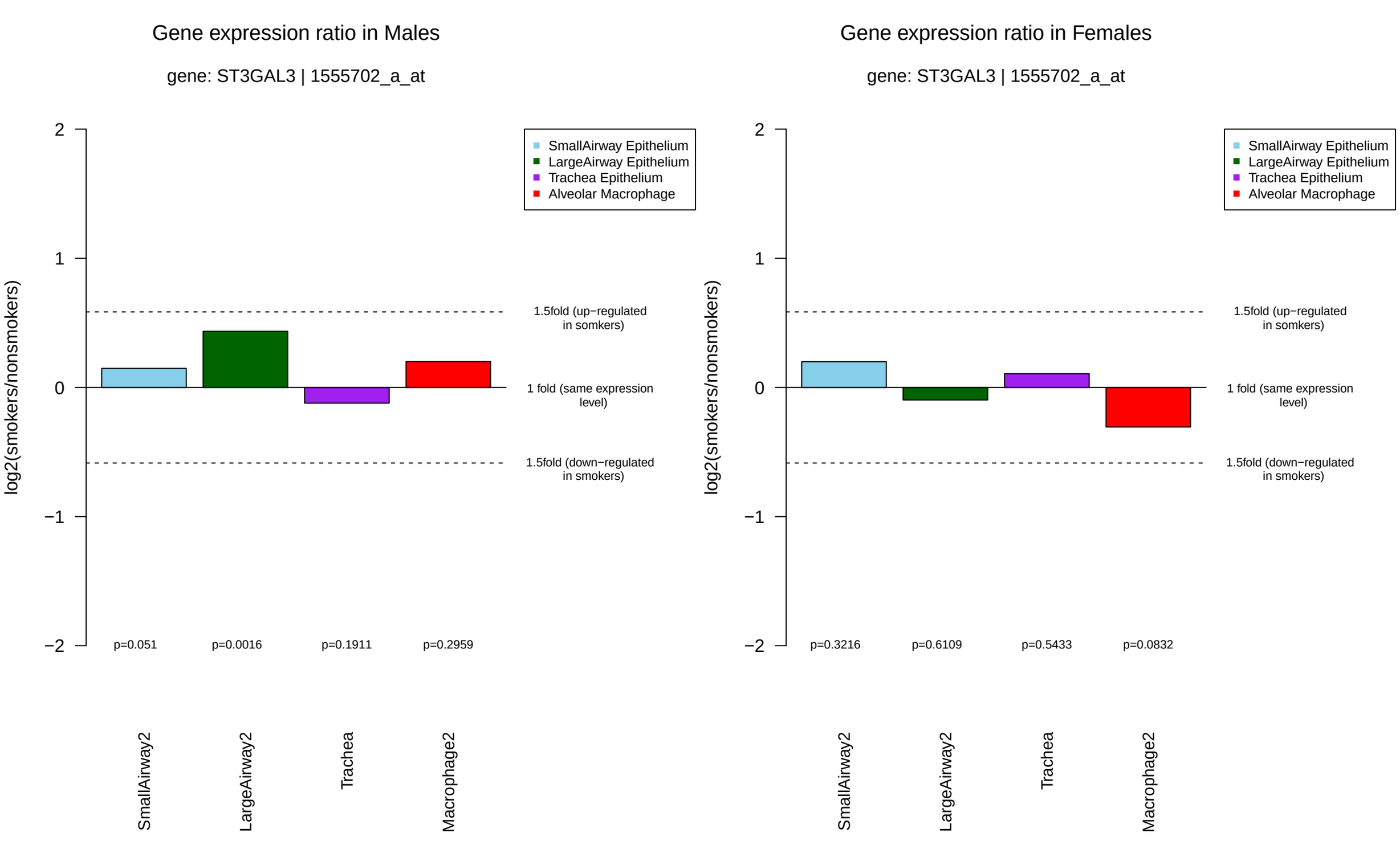
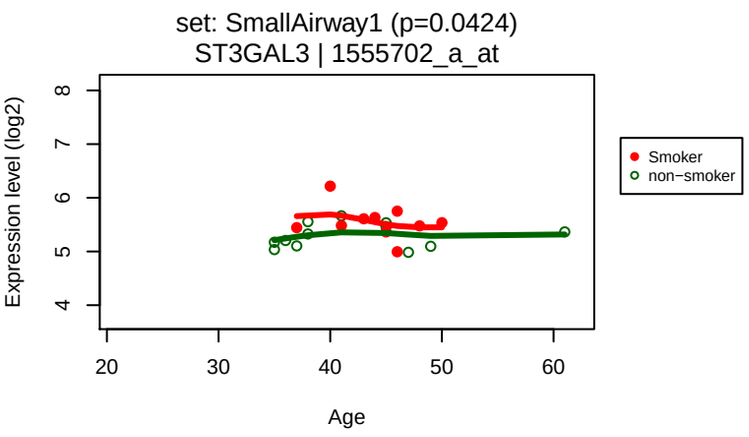
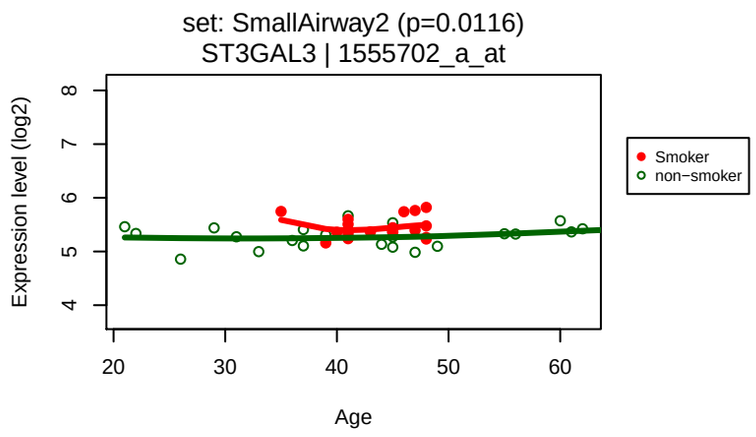
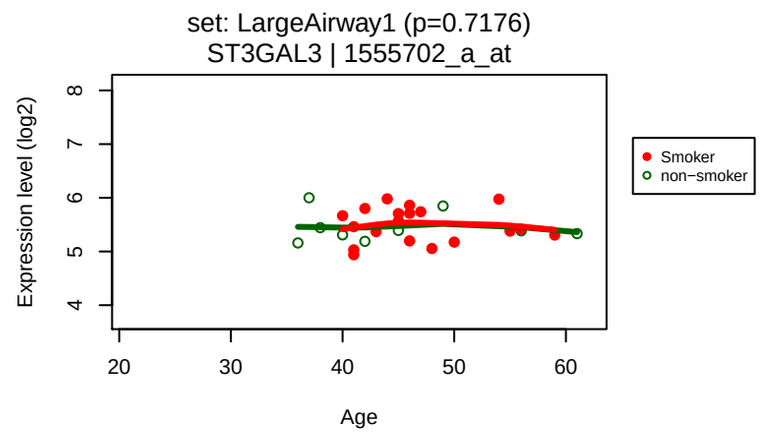

Gene expression of smokers and nonsmokers
gene: ST3GAL3 | 1555702_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0424
p=0.0116
p=0.0095
p=0.7176
p=0.0211
p=0.3377
p=0.8421
p=0.8374
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

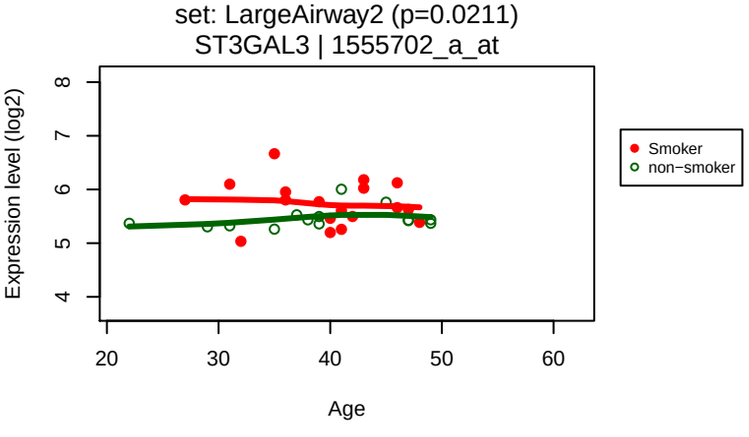

Gene expression ratio (smokers/nonsmokers)
gene: ST3GAL3 | 1555702_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0424
p=0.0116
p=0.0095
p=0.7176
p=0.0211
p=0.3377
p=0.8421
p=0.8374
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

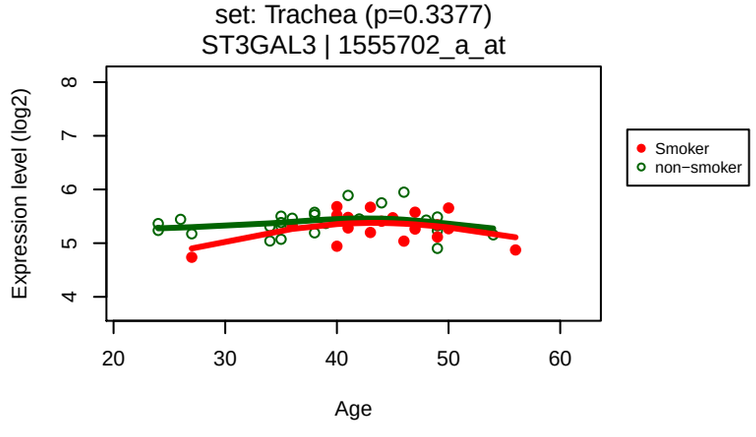

Gene expression ratio in Males
gene: ST3GAL3 | 1555702_a_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.051
p=0.0016
p=0.1911
p=0.2959
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ST3GAL3 | 1555702_a_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3216
p=0.6109
p=0.5433
p=0.0832
SmallAirway2
LargeAirway2
Trachea
Macrophage2

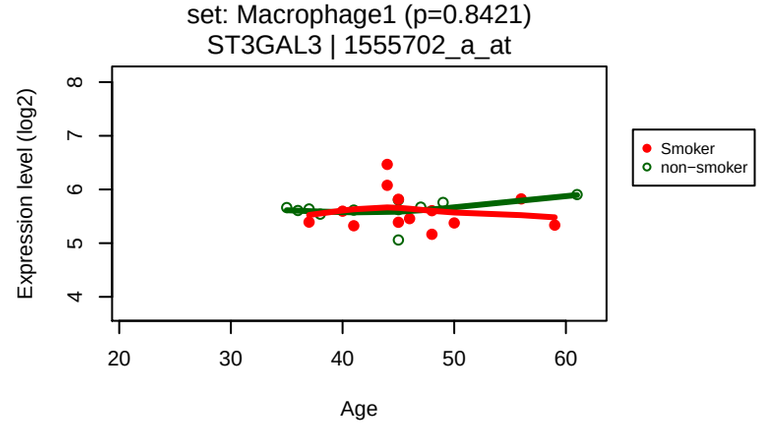

set: SmallAirway1 (p=0.0424)
ST3GAL3 | 1555702_a_at
Expression level (log2)
Age
Smoker
non−smoker

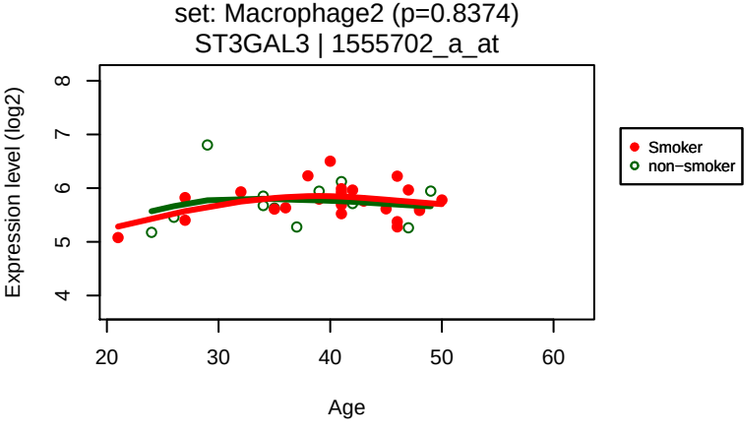

set: SmallAirway2 (p=0.0116)
ST3GAL3 | 1555702_a_at
Expression level (log2)
Age
Smoker
non−smoker

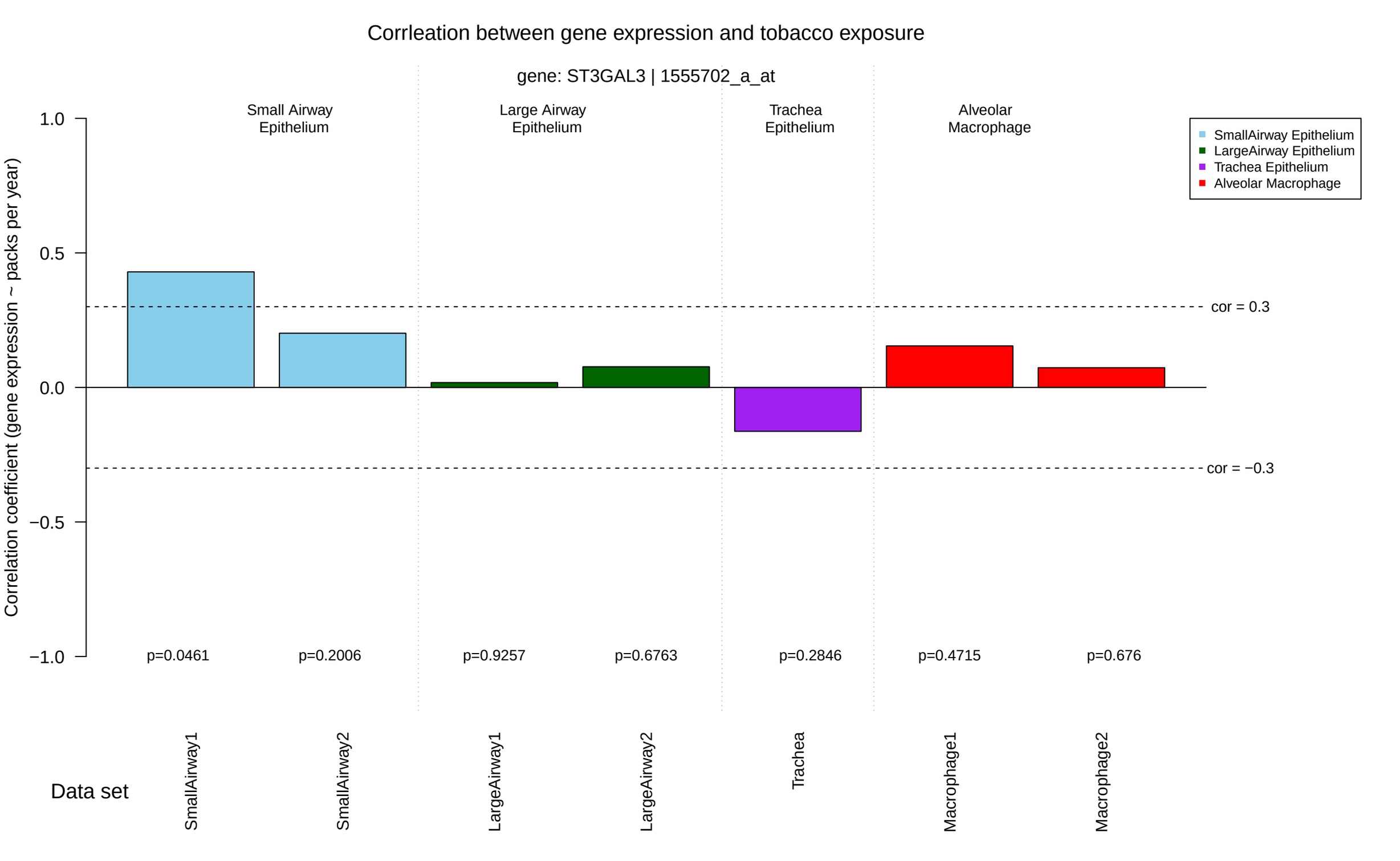

set: LargeAirway1 (p=0.7176)
ST3GAL3 | 1555702_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0211)
ST3GAL3 | 1555702_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.3377)
ST3GAL3 | 1555702_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8421)
ST3GAL3 | 1555702_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.8374)
ST3GAL3 | 1555702_a_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ST3GAL3 | 1555702_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0461
p=0.2006
p=0.9257
p=0.6763
p=0.2846
p=0.4715
p=0.676
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2